

Divanshu Sharma

Machine Learning Engineer

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SKILLS

Languages:	Python, C++, Bash
Libraries & Frameworks:	PyTorch, TensorFlow, Scikit-learn, Transformers, CNNs, GNNs, RAG, SFT, Hugging Face, LangChain, Whisper, DeBERTa, RoBERTa, PyTorch Geometric
MLOps & Cloud:	Docker, CI/CD (GitHub Actions), MLflow, Airflow, FastAPI, AWS (SageMaker, EC2), GCP, Streamlit
Databases:	PostgreSQL, MySQL, MongoDB, Vector (Qdrant, Chroma)

EXPERIENCE

Yale University | ML Researcher | Oct 2025 - Mar 2026

- Engineered a secure, open-source deep learning library from first principles (Python/NumPy, no external autograd), implementing a compiler-centric MPC architecture that delivers verifiable cryptographic security for privacy-preserving collaborative training.
- Designed a strided-convolution-based CNN and achieved 94.62% test accuracy on ChestMNIST (batch size 512, 5 epochs).

WorldQuant BRAIN | Research Consultant | Nov 2022 - July 2025

- Developed and backtested 15+ high-frequency alpha signals via Python and WorldQuant Simulation, achieving an average Sharpe Ratio of 1.8 - earning GOLD LEVEL (Top 1%) in the 2022 Global Alphathon.

PROJECTS & OPEN-SOURCE

End-to-End Indic ASR Optimization & Consensus Evaluation | [GitHub](#)

Technologies: Whisper, SpecAugment, Transformers, Beam Search, Devanagari NLP

- Achieved >48% absolute WER reduction on the FLEURS benchmark by fine-tuning Whisper-small with SpecAugment, Label Smoothing, and Cosine LR scheduling evaluated via a 5-beam Beam Search pipeline.
- Reduced transcription errors across **177,000** unique terms by building a Devanagari normalization engine and a phonetic reverse-transliteration layer, fusing 5 ASR models via Word-level Confusion Networks with majority-voting.

RenAissance OCR – Research-Grade Historical OCR Pipeline | [GitHub](#)

Technologies: CRAFT, SE-ResNet-BiLSTM CRNN, RoBERTa, Data Augmentation

- Reduced CER from 3.44% to 0.00% on a 183-sample validation set by chaining CRAFT text detection, an SE-ResNet-BiLSTM CRNN recognizer, and an LLM-based RoBERTa post-processor
- Scaled the pipeline to 12,792 training samples of degraded 16th–17th century Spanish texts using advanced data augmentation and a 7-step preprocessing workflow, lifting full-corpus Word Accuracy from 97.96% to 98.17%.

ML4SCI Task Solutions Repository (CMS)

Technologies: PyTorch, PyTorch Geometric, TensorFlow, CNN, GCN | [GitHub](#)

- Attained 80% AUC in electron-vs-photon classification by developing a custom ResNet-15 CNN on raw CMS detector data, directly improving high-energy physics event reconstruction quality.
- Cut training time by 30% - matching a complex GAT baseline's ROC curve - by implementing a Graph Convolutional Network for particle momentum regression; boosted quark/gluon jet classification by 7% via targeted VGG-style CNNs.

EDUCATION

University of Mumbai | Mumbai, India

Bachelor of Engineering, Computer Science | June 2025 | CGPA: 8/10.0

- Relevant Coursework: Machine Learning, Deep Learning, Data Structures & Algorithms, Probability & Statistics

ACHIEVEMENTS & CERTIFICATIONS

- GOLD LEVEL (Top 1 %), STAGE 1 Qualifier, WorldQuant BRAIN Global Alphathon 2022: Achieved a top-tier ranking among thousands of global participants by designing and submitting high-performing quantitative trading models.
- Earned Kaggle 2× Expert status; solved 3,000+ algorithmic problems across LeetCode (Knight), CodeChef (Max Rating 1662), and HackerRank.
- Completed ML Specialization, Deep Learning Specialization, TensorFlow Developer Certificate, and GANs Specialization (DeepLearning.AI, 2023–2024).